

Correction note

Prepended 21 September 2026 to the record as originally rendered, which follows unchanged.

Correction note — 21 September 2026 (D-42). The needs-review figures here were inflated by a classifier defect: API batches cut off at the token limit were cached as if the model had declined to classify them. At 442 rows the corrected figure is **128 (29.0%), not 183 (41.4%)**. This addendum explains the rise from 168/430 (39.1%) to 183/442 (41.4%) as newer titles drifting toward less classifiable strings. **That explanation is withdrawn:** it was never tested, and the rise is at least partly the defect. The addendum below is left as written. See `docs/d42_failure_record.md` for the cause, the evidence and the fix.

Network Visualiser — Day 17 Addendum

Items logged during Gate C

9 September 2026

Day 17 addendum — items logged during Gate C

Logged 9 September 2026. Everything below existed only in conversation until now. Gate C steps 1-3 passed before these were written; none of them block the gate.

This document is intended to be appended to `docs/project_state_decision_log.md`, which remains the canonical decision log.

Open defects

D-28b — graph artefact drift is undetected by design

`centrality.py` rebuilds the graph in memory via `graph.py`'s `load_nodes`, `choose_threshold` and `build_graph`, then writes `network_nodes.csv` itself. It never reads `graph.graphml`. So if `src.graph` is not re-run after an export change, `graph.graphml`, `similarity.npy`, `network_preview.png` and the tracked `graph_metrics.csv` silently describe a network that no longer exists, while every consumer downstream continues to work correctly.

This was found on 8 September with `graph_metrics.csv` at 430 nodes against a 442-row table, and cured by re-running `src.graph`. D-28 is closed. D-28b is the permanent fix: `centrality.py` should assert its node count against `graph_metrics.csv` and fail loudly on disagreement. Recurs on every export change until built.

Phase: P2.9.

D-29 — FINAL_PARAMS duplicated across projection modules

`projection.py:51` and `projection_3d.py:44` each carry `{"n_neighbors": 15, "min_dist": 0.1}`. Only `projection.py` runs the sweep, so the constant governing the rendered 3D canvas has no evidence attached to it and can drift from the one that does. Two-paths condition of the kind P2.8b exists to eliminate.

Phase: P2.9.

Related finding, not a defect. On the 442-row sweep, `15 / 0.1` scores 0.9474 and ranks fifth of twelve, behind `10 / 0.0` at 0.9595. The constant is retained deliberately. `min_dist=0.0` maximises trustworthiness by collapsing points onto each other, which is good for the metric and bad for a canvas meant to be read; `n_neighbors=10` buys local fidelity by discarding the global structure the product sells. Choosing 0.9474 for legibility is the trade, and it is stated here rather than left for a reviewer to find in the sweep file.

D-30 — matplotlib is an undeclared dependency

Imported at module level by `graph.py` (lines 38, 41), `projection.py` (line 32) and `clustering.py` (lines 138, 140), and absent from `requirements.txt`. A fresh clone cannot run the rebuild chain. `app.py` and `canvas.py` do not import it, so the deployed dashboard is unaffected. Resolved by D-31.

Phase: P2.9.

D-31 — no build/runtime manifest split

`requirements.txt` is a development manifest. Deploying from it installs `sentence-transformers` (and `torch`), `hdbscan`, `umap-learn`, `scikit-learn`, `pyvis`, `faker` and `pytest`, none of which `app.py` or `canvas.py` import at runtime — the dashboard reads `network_nodes.csv` and `projected_3d.csv`, and the model stack is build-time only. Very heavy cold start against a Community Cloud resource ceiling that has never been tested.

Fix: split into a build manifest and a runtime manifest, verified by importing `app.py` in a clean venv containing only the runtime set. This also disposes of D-30 cleanly — `matplotlib` goes in the build manifest and never reaches the deployed app.

Phase: blocks P2.9.

D-32 — the review queue is a fourth coverage surface

D-18 was scoped to three derivation sites: the summaries panel, the chat panel and the header. The review queue panel was never enumerated and kept the old phrasing. On a cold start it reads “183 of 183 still need review”, the exact stutter the header now suppresses when `n == m`; after a restore it correctly reads “5 of 183”, so only the equal-values branch is missing.

Fix after Gate C, together with a grep for any other `{n} of {m}` construction.

D-33 — the embeddings cache is keyed on row count, not content

`compute_embeddings` returns the cached matrix whenever `cached.shape[0] == len(people)`, and `main()` exposes no `recompute` flag. Edit a title without changing the row count and the stale 384-dimension vectors are reused through clustering, projection, graph and centrality, while the node table displays the new titles. It fails silently and looks correct.

Current mitigation is procedural: delete `data/cache/embeddings.npy` before any rebuild following an export change. The real fix is a content hash — noted in the module docstring as a Stage 2 upgrade.

Phase: P2.9. The procedural step must be in the README before then.

D-34 — stale comment block in app.py

`app.py:144–148` still narrates the pre-D-18 counter phrasing (“168 need review · 168 remaining”, “167 of 168 · 1 resolved”), which no longer matches the code shape below it. Cosmetic, but comment drift has caused at least two misreadings in this codebase, which is why it gets a number rather than a mention.

D-26 and D-27 — resolved on inspection

D-27 is closed. Commit `d5152b0` (8 September): the restore report merged corrections and notes into one fraction, hiding which kind was lost. A dropped correction returns a node to grey and costs one click from the review queue; a dropped note is the user’s own writing, gone. `sentence()` now reports each kind separately, and the drift block renders first, framed as kept. Verified live at Gate C step 2.

D-26 is closed, but its definition was lost in authoring. It appears in exactly one place — the summary line beneath the Gate B defect table — and in no commit message, code comment or ticket. The defect table itself runs D-11 to D-24 and has no row for it.

The probable cause is visible in `d5152b0`’s commit message, which contains a paragraph beginning mid-sentence: “was named for an invariant but asserted an exact string, so it broke on a rewording that did not violate it. It now checks the property, with a companion covering the opposite direction.” The opening line naming the subject is missing. That commit touches `tests/test_session_io.py`, where line 335 carries a matching comment, so D-26 was a brittle snapshot test asserting exact strings rather than the invariant it was named for — fixed in the same commit that closed D-25 and D-27.

Recorded as closed. The commit message is not amended: it is pushed, and a history rewrite is disproportionate to a lost sentence.

Process finding. This is the third defect caused by PowerShell file authoring in this project: D-20 was a BOM written by `Set-Content`, and this was truncation of a commit message written via a temp file. The `.NET WriteAllText` with `UTF8Encoding($false)` rule addresses encoding but not truncation. Commit messages should be read back before committing — `git log -1 --format=%B` — whenever the message is authored via a temp file rather than typed inline.

Untested execution paths

Not defects. Recorded so the gate log is honest about what was never exercised.

MISS_TITLE_CHANGED and the mixed-kind restore render. Gate C step 2 exercised `MISS_NO_MATCH` on a note only, and step 13 cannot reach the other branch because titles cannot move inside a single session. First real execution will be a user’s second upload against a changed export. Cheap pre-flight: a copied session JSON with one entry’s `role` altered — no rebuild, no API spend.

SidecarMismatch. Raised by the loader when the 3D sidecar disagrees with the base table. Has never fired.

Backlog

P3.12 — re-import correction churn

Between the July and September exports, 114 of 430 people changed their headline, several across SOC major group boundaries (“Head of Corporate Communications” to “Sales Executive”; “Security Supervisor” to “Internship”). Corrections key on `name|role`, so every one of those returns as `MISS_TITLE_CHANGED` and goes back to grey. That is correct — a verdict about the old title has no authority over the new one — but it means a user pays the review tax again on people they have already reviewed, every time they re-upload.

The backlog also grew faster than the network: 168/430 (39.1%) became 183/442 (41.4%), because the new titles drift toward less classifiable strings — “Internship”, “Graduate Trainee”, “Industrial Placement”, “Ride Leader”. Input drift, not classifier decay; `reclassified: 0` rules out the latter.

Candidate mitigation: when a title changes but the person is the same, offer to carry the prior correction forward as a *suggestion* the user confirms, never as a silent re-application.

Phase: 3, alongside P3.10.

P3.13 — chat history in the session file

Chat exchanges are lost on refresh. The session file persists corrections, notes and About text but not the conversation, so a user who asks a good question and closes the tab loses the answer.

Deferred to Phase 3 rather than built now: the mechanism is a P2.8c extension, but its value depends on P3.1 team composition answers, which are the content worth keeping. Building it against search-grade answers would under-specify it.

Design question to resolve before implementation. The session file currently holds only what the user wrote. Chat history holds generated prose naming connections, which changes the disclosure obligation on the download panel. Preferred option: an explicit opt-in checkbox at download, keeping the default file as honest as it is today. Alternatives are inclusion by default with an expanded warning, or a separate file.

Phase: 3.

Gate C — verdict

Closed 9 September 2026 at 13 of 13. One continuous browser session, no restart, no crash, on the 442-row table.

Step	Verdict
1 Cold start — four surfaces agree	PASS
2 Restore, banner reports by kind	PASS
3 Live search on keystroke	PASS
4 Search by SOC group name	PASS
5 Zero-match empty state	PASS
6 Clear via ×	PASS
7 Needs-review card renders	PASS
8 Assign a group, live update	PASS
9 Review queue skip and assign	PASS
10 Note, tagged and searchable	PASS
11 About text and privacy note	PASS
12 Grounded chat, caption still agrees	PASS
13 Round trip, idempotent	PASS

Gate B verified features in isolation. Gate C verified they compose across a session carrying two corrections, a not-an-occupation, a skip, a note, About text and a chat exchange, with all four coverage surfaces still agreeing at the end.

Executed live for the first time in this gate: both branches of the D-18 counter wording, the D-12 nonce-bump clear, the D-23 bordered restore container, `STATE_NOT_OCCUPATION` from the queue surface, the empty-set `match_indices` sentinel, notes reaching `filter_nodes`, and a restore-of-a-restore round trip.

Caveat on step 2. It was verified against a temporary 441-row table created by removing one connection from the export to exercise `MISS_NO_MATCH`. The export was subsequently restored and the chain rebuilt to 442, reproducing the

graph exactly. The step-2 state is not reproducible from the current artefacts.

Methodology findings

Two claims the write-up makes that had never been tested, both evidenced on 9 September.

End-to-end determinism

`projected_3d.csv` and a copy generated by a separate run of `src.projection_3d` returned identical SHA-256 hashes, and `graph_metrics.csv` came back byte-identical across a full delete-and-rebuild cycle (embeddings deleted, whole chain re-run) at $\tau=0.6258$, 1769 edges, 86 isolates. `random_state=SEED` reproduces the layout across runs, not merely within one.

Classification stability across rebuilds

`tools/diff_rebuild.py` compared the 430-row and 442-row tables keyed on `(name, role)`: 316 people present in both with unchanged name and title, of whom **0 were reclassified**. The drop from 262 to 259 classified is fully explained by departures, not by non-determinism in the Claude classifier. Corrections made against an earlier export remain trustworthy. Sampling confirmed the 114 title moves are genuine headline edits, not whitespace, case or truncation artefacts, so the instrument is sound.

Graph metrics, 430 to 442

Metric	430	442
n_edges	1720	1769
threshold_tau	0.6205	0.6258
mean_degree	8.0	8.005
n_isolates	82 (19.1%)	86 (19.5%)
n_components	102	104
modularity_hdbscan	0.3352	0.3429
modularity_greedy	0.6731	0.7144

The isolate count is proportionally stable — the earlier 19.5% flag was not a regression. Mean degree held to target. The eigenvector diagnostic reports 167/442 nodes collapsing to approximately zero, re-earning the PageRank decision on the current table rather than inheriting it from Day 7.

*Data attribution: this project includes information from the O*NET 30.3 Database by the U.S. Department of Labor, Employment and Training Administration (USDOL/ETA), used under the CC BY 4.0 license. O*NET® is a trademark of USDOL/ETA.*